

3.14 Structured Probabilistic Models

Machine learning algorithms often involve probability distributions over a very large number of random variables. Often, these probability distributions involve direct interactions between relatively few variables. Using a single function to describe the entire joint probability distribution can be very inefficient (both computationally and statistically).

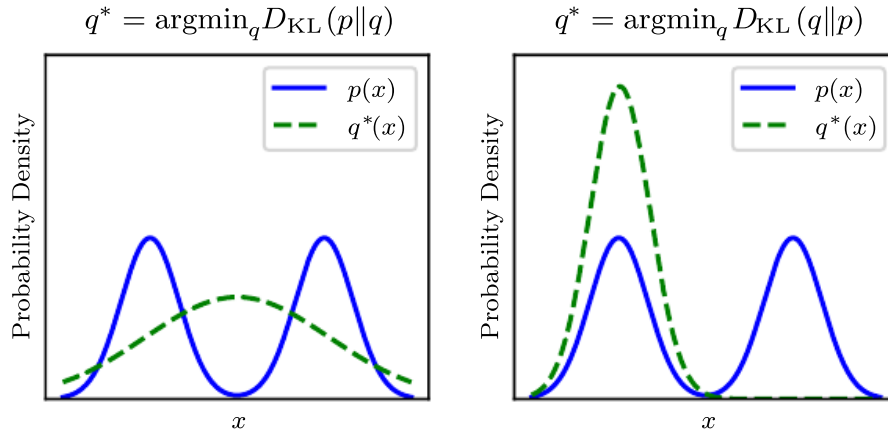


Figure 3.6: The KL divergence is asymmetric. Suppose we have a distribution $p(x)$ and wish to approximate it with another distribution $q(x)$. We have the choice of minimizing either $D_{\text{KL}}(p||q)$ or $D_{\text{KL}}(q||p)$. We illustrate the effect of this choice using a mixture of two Gaussians for p , and a single Gaussian for q . The choice of which direction of the KL divergence to use is problem dependent. Some applications require an approximation that usually places high probability anywhere that the true distribution places high probability, while other applications require an approximation that rarely places high probability anywhere that the true distribution places low probability. The choice of the direction of the KL divergence reflects which of these considerations takes priority for each application. *(Left)* The effect of minimizing $D_{\text{KL}}(p||q)$. In this case, we select a q that has high probability where p has high probability. When p has multiple modes, q chooses to blur the modes together, in order to put high probability mass on all of them. *(Right)* The effect of minimizing $D_{\text{KL}}(q||p)$. In this case, we select a q that has low probability where p has low probability. When p has multiple modes that are sufficiently widely separated, as in this figure, the KL divergence is minimized by choosing a single mode, to avoid putting probability mass in the low-probability areas between modes of p . Here, we illustrate the outcome when q is chosen to emphasize the left mode. We could also have achieved an equal value of the KL divergence by choosing the right mode. If the modes are not separated by a sufficiently strong low-probability region, then this direction of the KL divergence can still choose to blur the modes.

Instead of using a single function to represent a probability distribution, we can split a probability distribution into many factors that we multiply together. For example, suppose we have three random variables: a , b and c . Suppose that a influences the value of b , and b influences the value of c , but that a and c are independent given b . We can represent the probability distribution over all three variables as a product of probability distributions over two variables:

$$p(a, b, c) = p(a)p(b | a)p(c | b). \quad (3.52)$$

These factorizations can greatly reduce the number of parameters needed to describe the distribution. Each factor uses a number of parameters that is exponential in the number of variables in the factor. This means that we can greatly reduce the cost of representing a distribution if we are able to find a factorization into distributions over fewer variables.

We can describe these kinds of factorizations using graphs. Here, we use the word “graph” in the sense of graph theory: a set of vertices that may be connected to each other with edges. When we represent the factorization of a probability distribution with a graph, we call it a **structured probabilistic model**, or **graphical model**.

There are two main kinds of structured probabilistic models: directed and undirected. Both kinds of graphical models use a graph $\mathcal{G}$ in which each node in the graph corresponds to a random variable, and an edge connecting two random variables means that the probability distribution is able to represent direct interactions between those two random variables.

Directed models use graphs with directed edges, and they represent factorizations into conditional probability distributions, as in the example above. Specifically, a directed model contains one factor for every random variable x_i in the distribution, and that factor consists of the conditional distribution over x_i given the parents of x_i , denoted $Pa_{\mathcal{G}}(x_i)$:

$$p(\mathbf{x}) = \prod_i p(x_i | Pa_{\mathcal{G}}(x_i)). \quad (3.53)$$

See figure 3.7 for an example of a directed graph and the factorization of probability distributions it represents.

Undirected models use graphs with undirected edges, and they represent factorizations into a set of functions; unlike in the directed case, these functions are usually not probability distributions of any kind. Any set of nodes that are all connected to each other in $\mathcal{G}$ is called a clique. Each clique $\mathcal{C}^{(i)}$ in an undirected model is associated with a factor $\phi^{(i)}(\mathcal{C}^{(i)})$. These factors are just functions, not

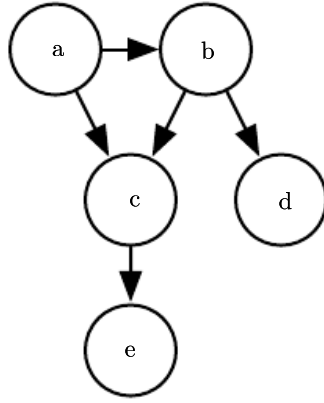


Figure 3.7: A directed graphical model over random variables a , b , c , d and e . This graph corresponds to probability distributions that can be factored as

$$p(a, b, c, d, e) = p(a)p(b | a)p(c | a, b)p(d | b)p(e | c). \quad (3.54)$$

This graphical model enables us to quickly see some properties of the distribution. For example, a and c interact directly, but a and e interact only indirectly via c .

probability distributions. The output of each factor must be non-negative, but there is no constraint that the factor must sum or integrate to 1 like a probability distribution.

The probability of a configuration of random variables is **proportional** to the product of all these factors—assignments that result in larger factor values are more likely. Of course, there is no guarantee that this product will sum to 1. We therefore divide by a normalizing constant Z , defined to be the sum or integral over all states of the product of the ϕ functions, in order to obtain a normalized probability distribution:

$$p(\mathbf{x}) = \frac{1}{Z} \prod_i \phi^{(i)}(\mathcal{C}^{(i)}). \quad (3.55)$$

See figure 3.8 for an example of an undirected graph and the factorization of probability distributions it represents.

Keep in mind that these graphical representations of factorizations are a language for describing probability distributions. They are not mutually exclusive families of probability distributions. Being directed or undirected is not a property of a probability distribution; it is a property of a particular **description** of a probability distribution, but any probability distribution may be described in both ways.

Throughout parts I and II of this book, we use structured probabilistic models

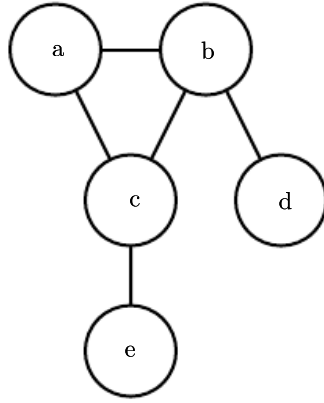


Figure 3.8: An undirected graphical model over random variables a , b , c , d and e . This graph corresponds to probability distributions that can be factored as

$$p(a, b, c, d, e) = \frac{1}{Z} \phi^{(1)}(a, b, c) \phi^{(2)}(b, d) \phi^{(3)}(c, e). \quad (3.56)$$

This graphical model enables us to quickly see some properties of the distribution. For example, a and c interact directly, but a and e interact only indirectly via c .

merely as a language to describe which direct probabilistic relationships different machine learning algorithms choose to represent. No further understanding of structured probabilistic models is needed until the discussion of research topics, in part [III](#), where we explore structured probabilistic models in much greater detail.

This chapter has reviewed the basic concepts of probability theory that are most relevant to deep learning. One more set of fundamental mathematical tools remains: numerical methods.